APPENDIX S2 – Overview of DNA extraction, library preparation, target enrichment, and Illumina Sequencing

These steps are an abridged overview of the laboratory protocols based on a previous phylogenomics study on *Myricaria* (Weitkämper, K., Lyakh, E. M., Morales-Briones, D. F., Weitkämper, F., and Werth, S., under review).

DNA extraction

- A modified CTAB extraction protocol (Doyle and Doyle, 1987) used to extract DNA from the selected specimens. This protocol was modified to maximise yield:
    - One to two branchlets of each specimen ground to powder using a TissueLyser II (Qiagen, Venlo, Netherlands) bead mill
    - 8µL of β-mercaptoethanol added to each sample in addition to 500µL CTAB before incubation (Doyle and Doyle, 1987; steps 1 and 2).
    - Extraction with 500µL chloroformisoamyl alcohol (24:1) performed twice (steps 3 and 4).
    - -20 °C 300µL 70% isopropanol added, then samples chilled overnight at -20 °C to maximise DNA precipitation.
    - Supernatant removed and the resulting DNA pellet redissolved in DNA-free purified water.
- Measured DNA concentration of extractions with the dsDNA HS DNA Assay Kit on a Qubit 4 Fluorometer (Thermo Fisher Scientific, Waltham, Massachusetts, USA).
- Ran Agarose gel electrophoresis (1% gels) stained with ethidium bromide to assess the DNA fragment lengths. Most samples had short fragment lengths (gel streaks were mostly under 500 bp), so sonication was not necessary, but longer fragment lengths sonicated for 60s or 80s in a 50µL Covaris microTUBE using a Covaris M220 sonicator (Covaris, Woburn, MA, USA). The settings used were: peak power 75.0 W; duty cycle at 10.0%; cycles/burst at 200 (Covaris, 2024).
- Assessed DNA purity using a NanoDrop spectrophotometer (Thermo Fisher Scientific). Low-quality extracts cleaned with SPRISelect or AMPure XP beads (both by Beckman Coulter, Brea, CA, USA) with 1:1.8 DNA extract to bead volume ratio.

Library preparation and pooling

- DNA libraries made with the NEBNext Ultra II DNA Library Preparation Kit (New England Biolabs, Ipswich, MA, USA), at half volume of manufacturer's protocol (New England Biolabs, 2022) to reduce costs.
- Adaptor was diluted one part in ten in nuclease-free water.
- To size select adaptor-ligated DNA, 12.5µL SPRISelect beads were added at step 3A.2 of the protocol and 22.5µL beads were added at step 3A.6. Library concentration measured with a Qubit 4, checked on a TapeStation with HS D1000 Screen Tape (Agilent, Santa Clara, CA, USA).
- Libraries normalised by molarity, then pooled by average fragment size into four pools containing 22–24 libraries. Pools dehydrated, then eluted in 7.5µL nuclease-free water, resulting in roughly 1µg DNA in each pool.

Target capture and Illumina sequencing

- Target capture carried out following the Standard Plants myBaits protocol v.5.2 (Daicel Arbor Biosciences, 2022) for the Angiosperms353 baits set, with hybridisation and wash temperatures set to 60°C.
- Incubated hybridisation reaction (Part 1) for 20 hours at 60°C (same as wash temperature).

- The final PCR amplification (Part 3) performed using NEBNext Ultra II Q5 Polymerase (New England Biolabs) with myBaits beads (Daicel Arbor Biosciences, Ann Arbor, MI, USA) in solution, for 14 cycles.
- Target-enriched library pools cleaned using the NucleoSpin Gel and PCR Clean-up kit (Macherey-Nagel, Düren, Germany) following protocol 5.1 PCR Cleanup (Macherey-Nagel, 2017).
- The pools quantified again using Qubit and TapeStation, then pooled together for two final pools with molarity of roughly 7nm at 30µL each.
- Final pooled libraries sequenced by the Core Genomics Facility of the Helmholtz Zentrum Munich, Germany, on Illumina NextSeq 1000 (Illumina, San Diego, CA, USA) with 150 bp paired-end reads on two P1.300 flow cells, with an average of 1,983,379 reads per sample or library.

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